

SN Systems Software Ltd



"Maximum effects, no side-effects"



John Roberts

Gil Jaysmith

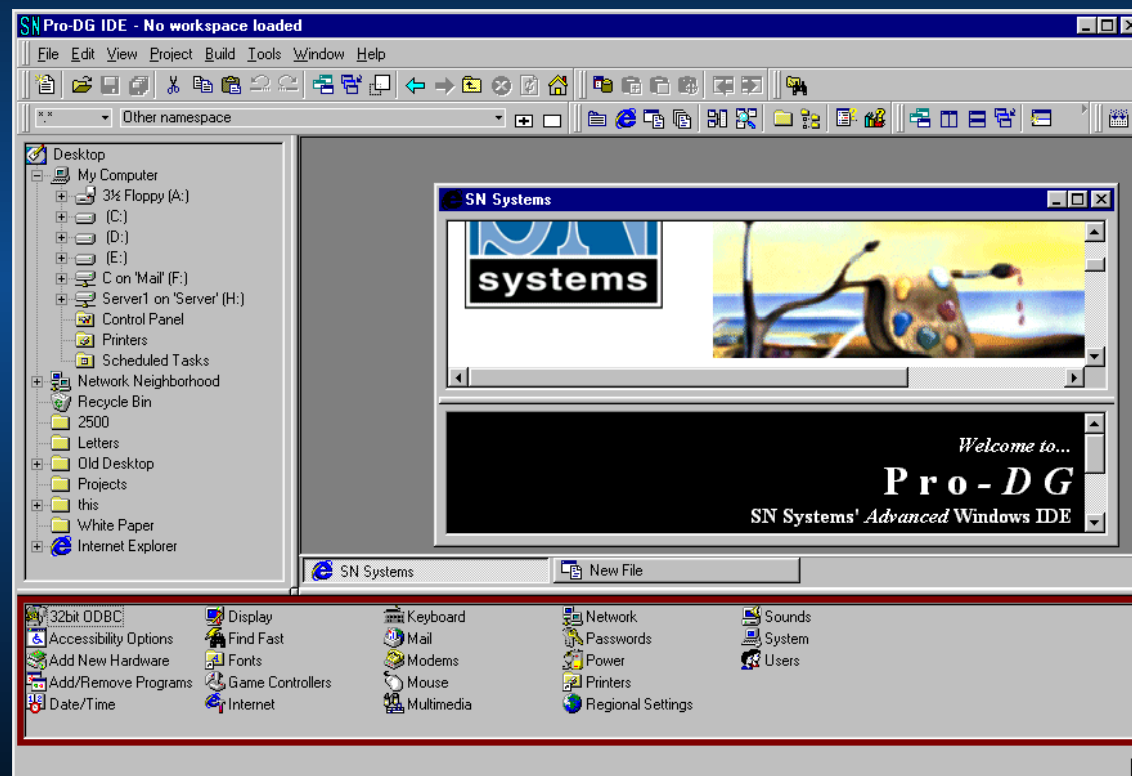
- Aims
- Visual Technology
- Build Manager
- Editor
- Shell Integration
- Customisation
- Extensibility
- How it fits with existing tools

Aims of Pro-DG

- Simple to install
- Provide single environment for all our tools
- Enhance “View & Pane” to next generation
- Make game-building easier
- “Integrate with everything”
- Easy to maintain, configure, extend
- Shorten our time-to-release for new features

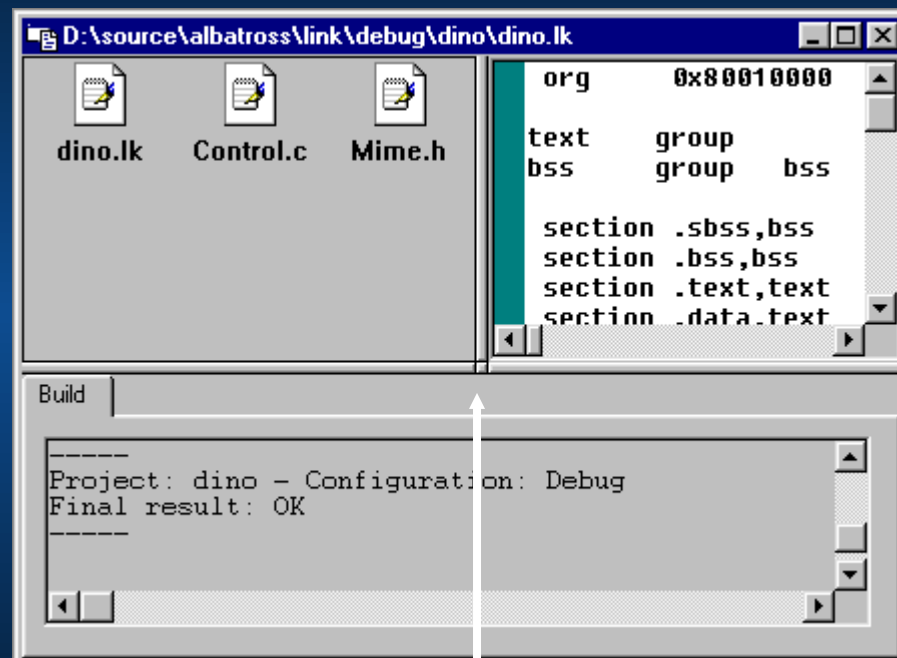
Visual Technology

- “View and Pane”
- the sequel!
- Style established
in DOS and
Windows
debuggers
- This is the next
generation



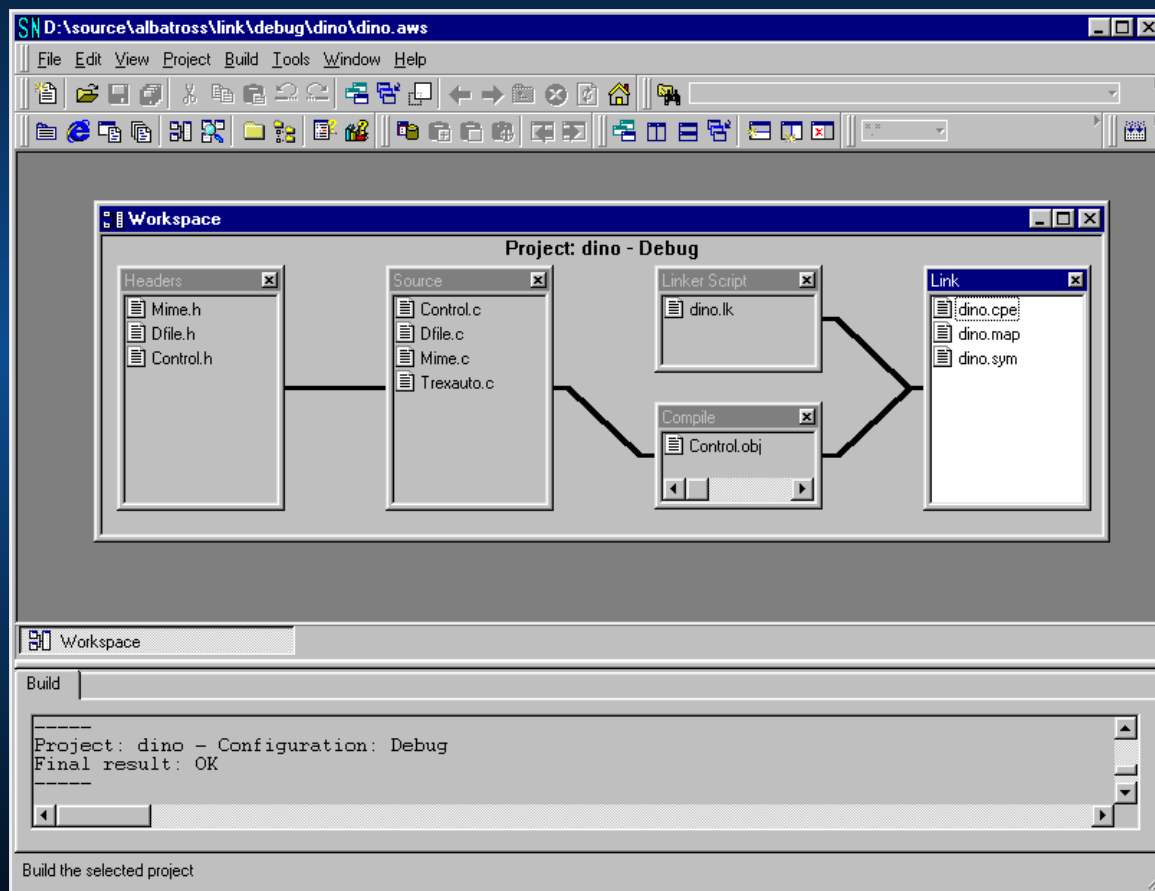
Views and Panes

- Each view is an MDI child window
- Each view can have one or more panes
- Any single pane can be zoomed to occupy its view
- Resize using T-junctions and cross-junctions



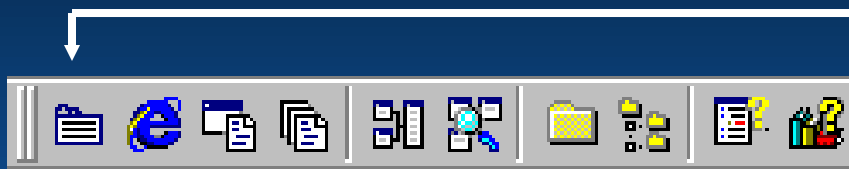
Frame Panes

- Like dockable toolbars
- Client area reduced
- Always visible



Pane Targeting

➤ Pane Targeting toolbar



➤ Pick a pane type, drop it where you want

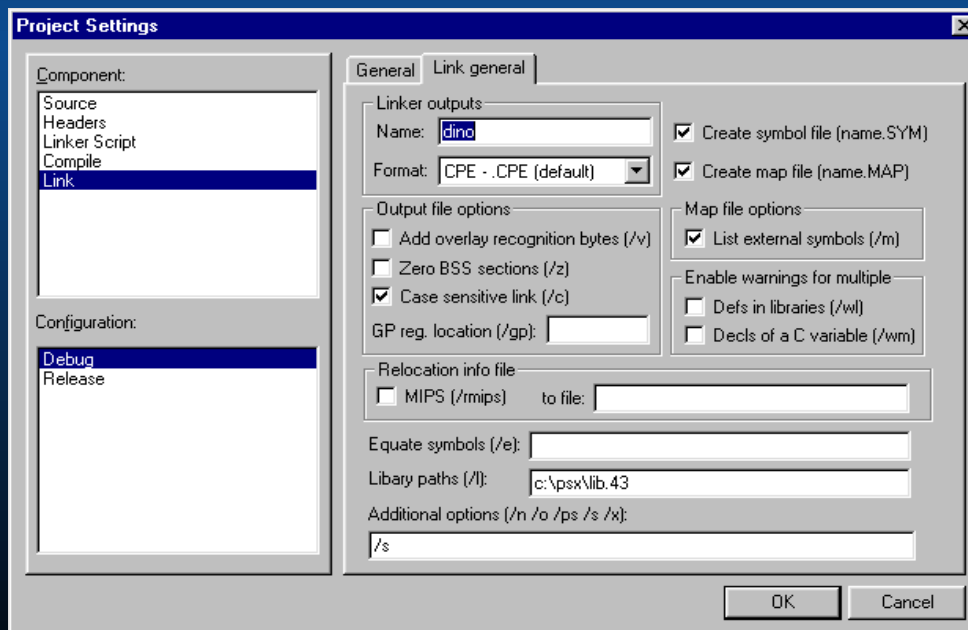
- Create new View
- Override existing pane
- Split pane in two
- Place in frame

Output
Explorer
Editor
Source Manager
Project
Folders
Shell Tree
Shell Folder
HTML Help Contents
HTML Help Content

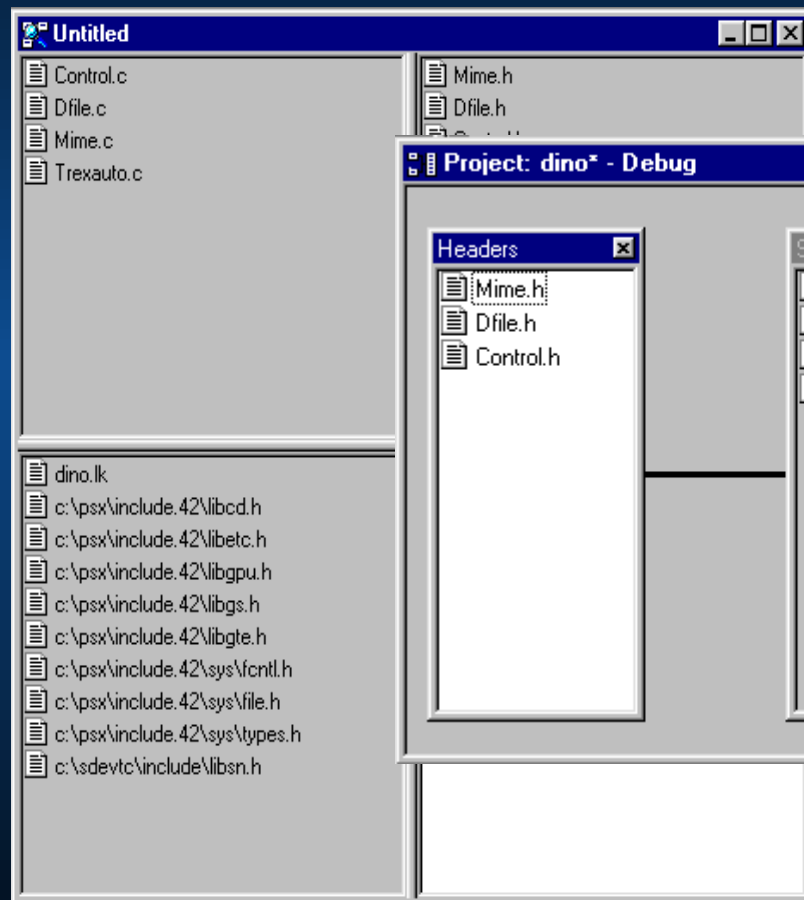
Pane History

- Each pane site remembers the types of panes you've had there
- Switch back and forth between them
- History includes pane's state, e.g.
 - which file was being edited
 - which web page was being browsed

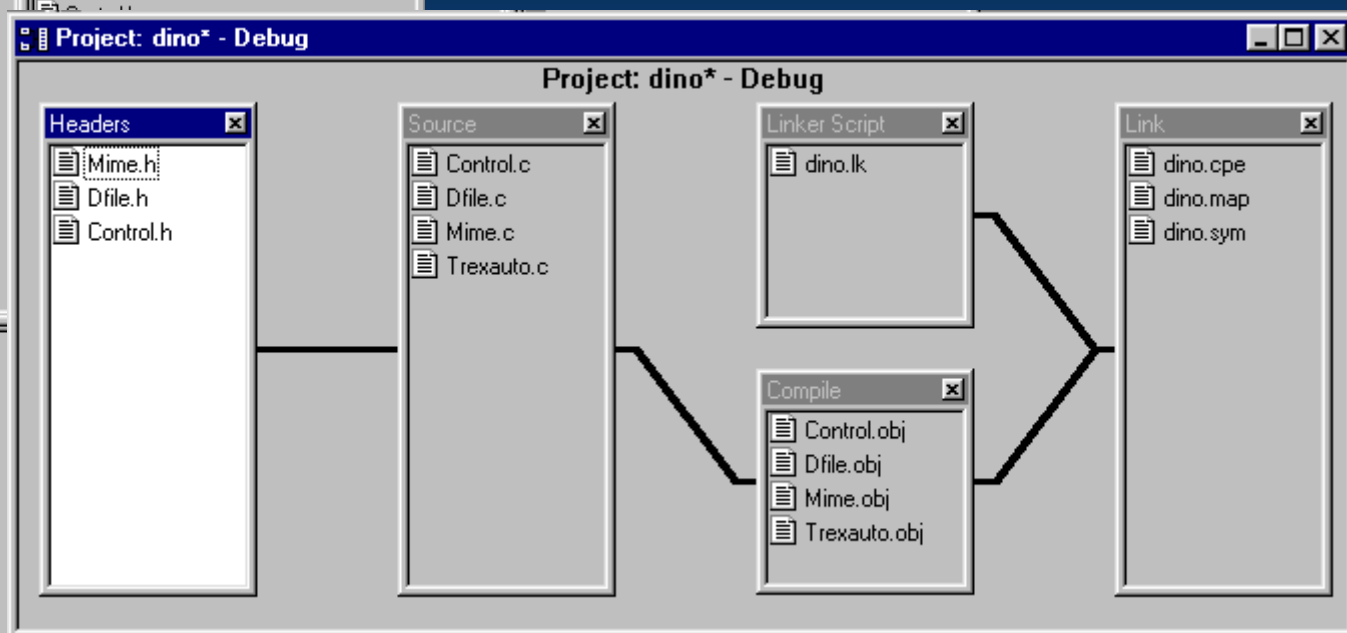
- Visualise your files and build processes
- Multiple configurations
- Configure settings all at once



Build Manager Panes

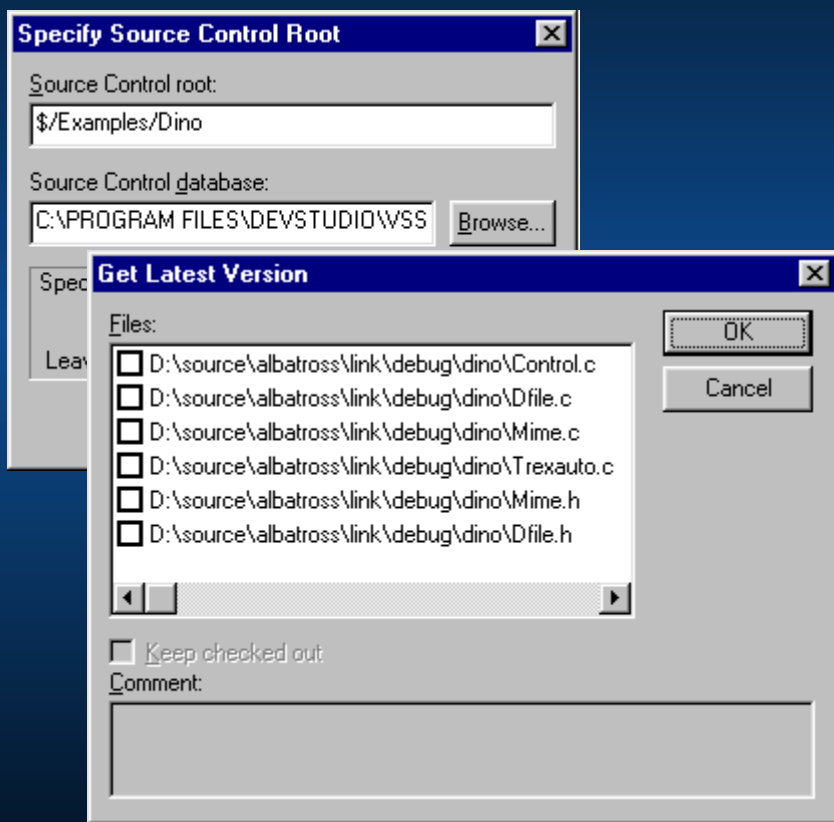


➤ Folder view...



➤ ... Or Graphical View

Source Control



- First: add project to Source Control
- Then: check files out when you need to work with them
- Works with Microsoft SourceSafe v5.0
- Other SCC systems to follow (e.g. PVCS)

```

D:\source\albatross\link\debug\dino\Trexauto.c

static int lf[16] = { 1,1,1,1,1,0,0,0,0,0,0,0,0,0,0,0 };
static int rf[16] = { 0,0,0,0,0,0,0,0,1,1,1,1,1,0,0,0 };
static int ln[16] = { 0,0,0,0,1,1,1,0,0,0,0,0,0,0,0,0 };
static int rn[16] = { 0,0,0,0,0,0,0,0,0,0,0,0,1,1,1,0 };

int alpha;
int modeflag;
int modeflagex;
u_long fn;                /* frame (field) No. */

u_long *SCRATCH = (u_long *)0x1f800000;

/*****
#ifdef CDFS
#define DFILENO 3

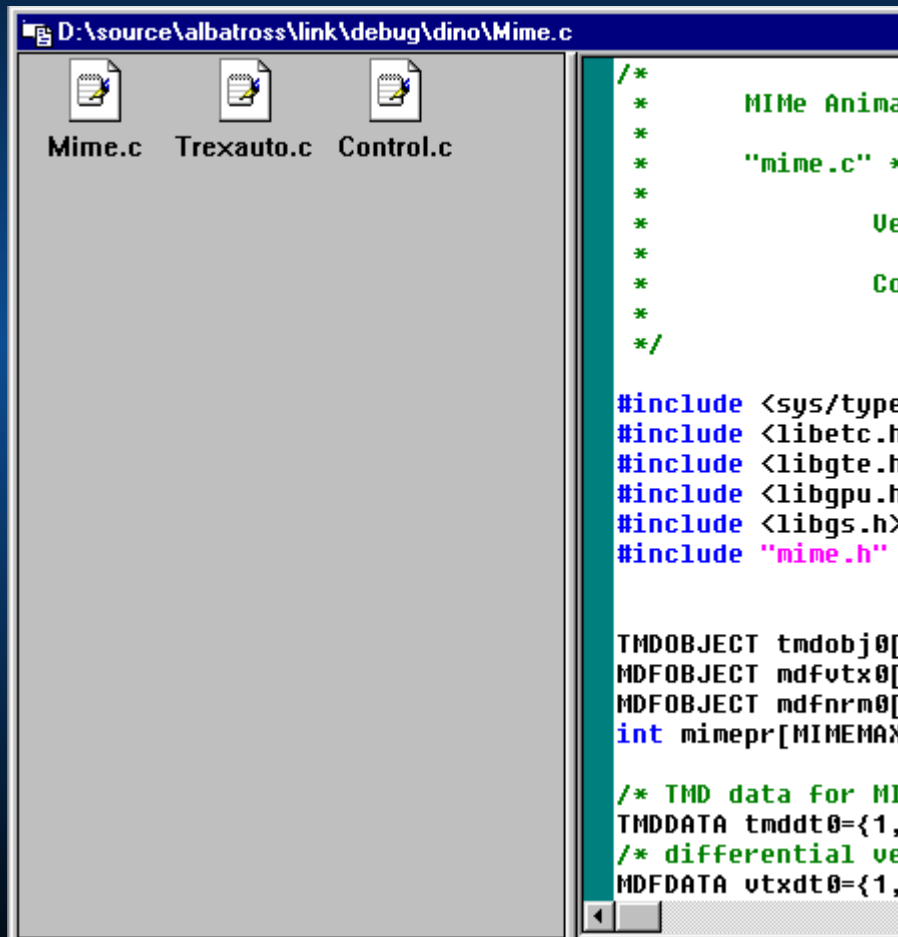
static FILE_INFO dfile[DFILENO] = {
    { "\\DINO\\DINOBASE.TMD;1", (void *)0x800a0000, 0 },
    { "\\DINO\\DINOBASE.VDF;1", (void *)0x80080000, 0 },
    { "\\DINO\\DINO.TIM;1", (void *)0x80100000, 0 }
};
#endif

/***** MAIN START *****/
main()
{
    int i;
    GsDOBJ5 *op;           /* pointer of Object Handler
    /* IIUWFFNngnnhüüi|C^ */

```

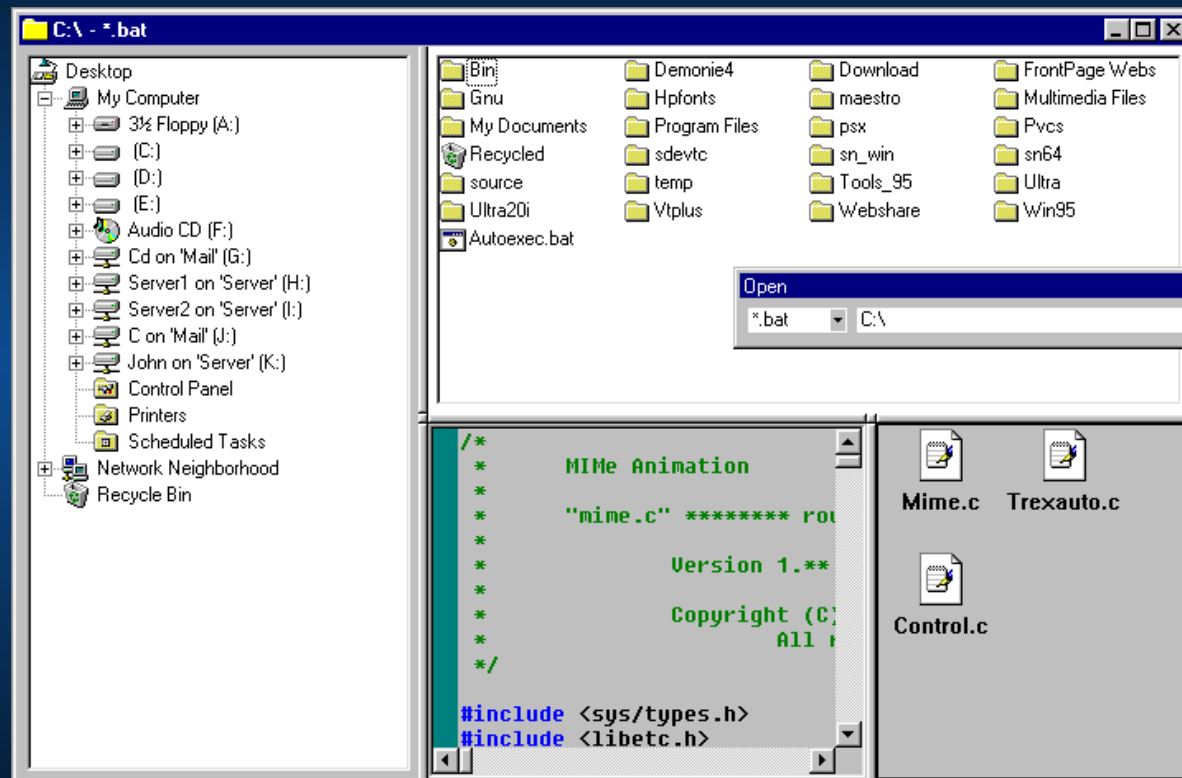
- It's an editor!
- Language-aware - C and C++
- Syntax colouring
- Indentation
- Undo/Redo
- Knows about Source Control
- Necessary for the future

Source Manager



- Buffers all the files you've edited
- Reopen any recent file from here

Shell Integration



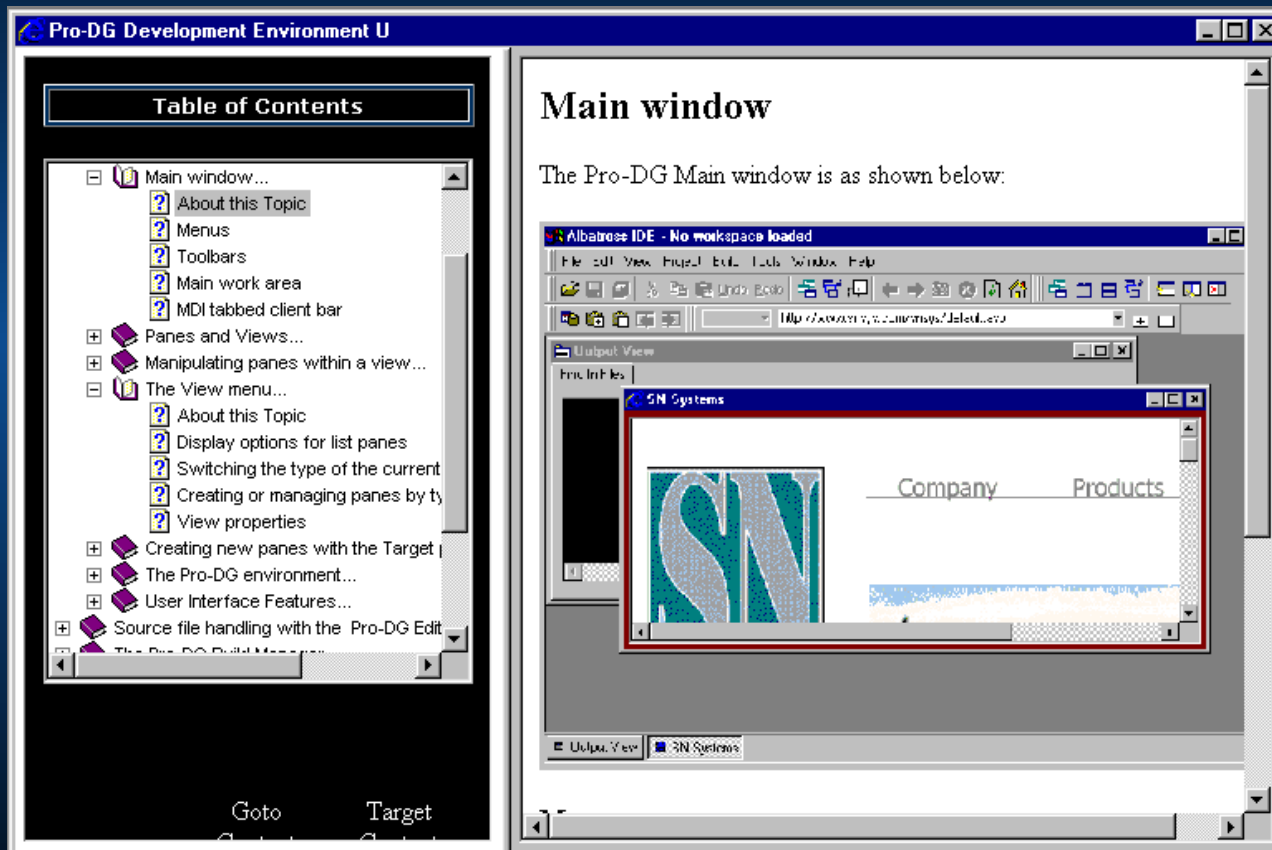
- Two pane types: Shell Tree and Shell Folder
- Adjacent Tree and Folder panes tied to create Windows Explorer lookalike
- Filter the folder contents

Explorer Pane



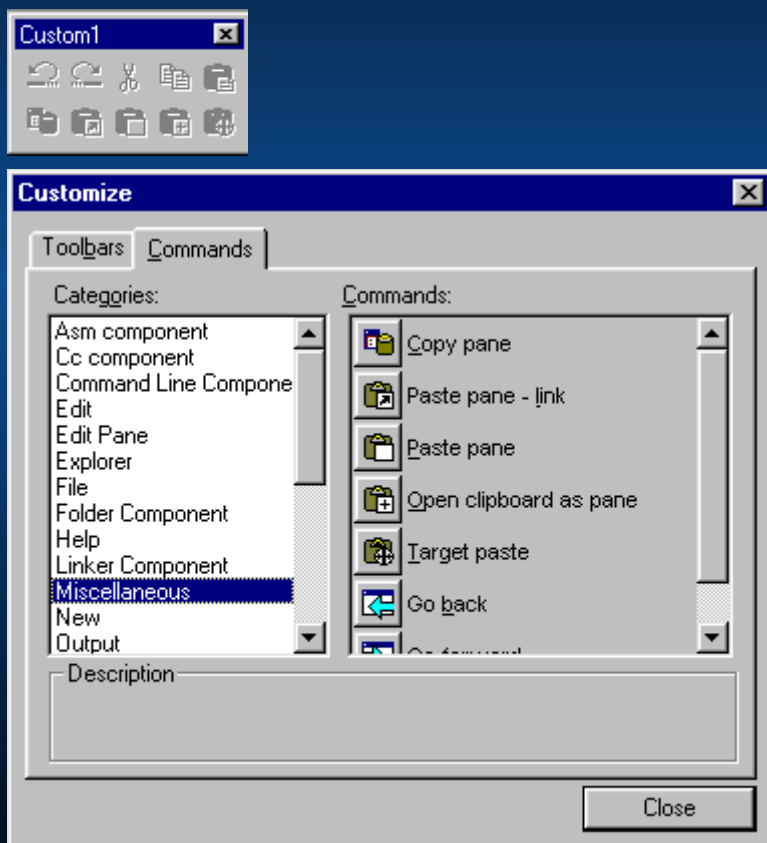
- Uses WebBrowser control to display HTML
- Can also display Dynamic HTML
- Hosts our help system!

Developer Conference 98 Online Help



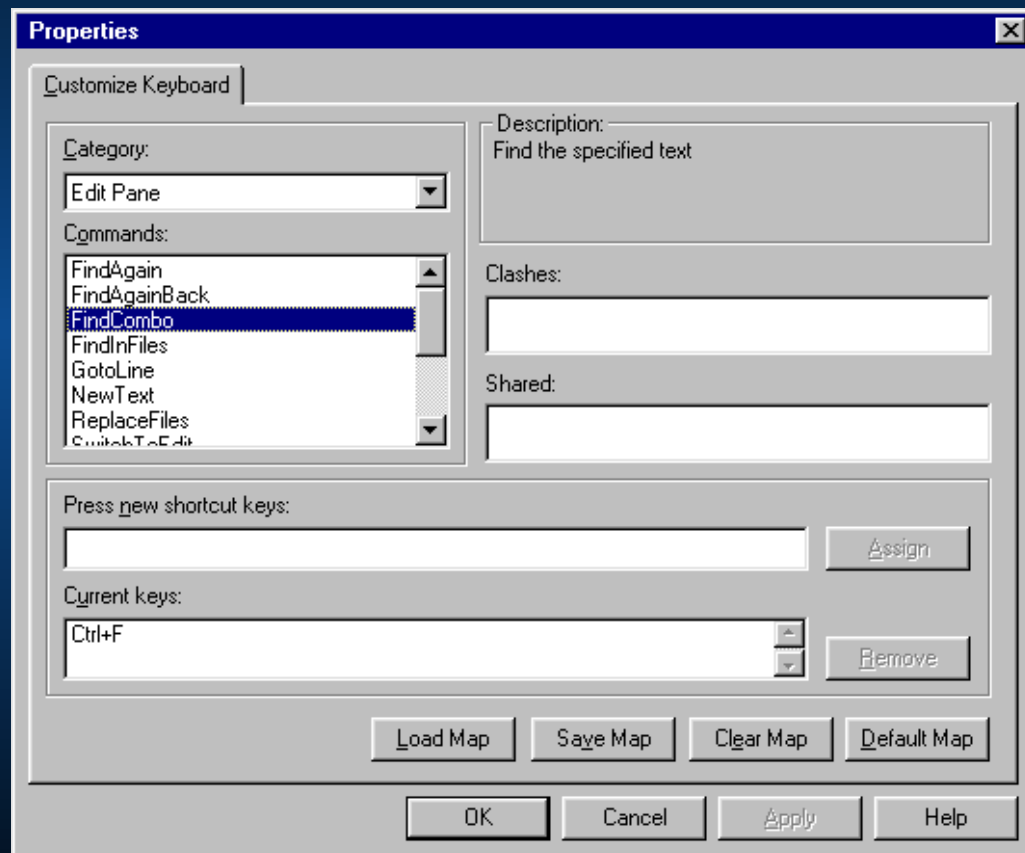
- HTML Help system with synchronised Contents and Content panes
- To be updated by SN with each software version

Customising Toolbars & Menus

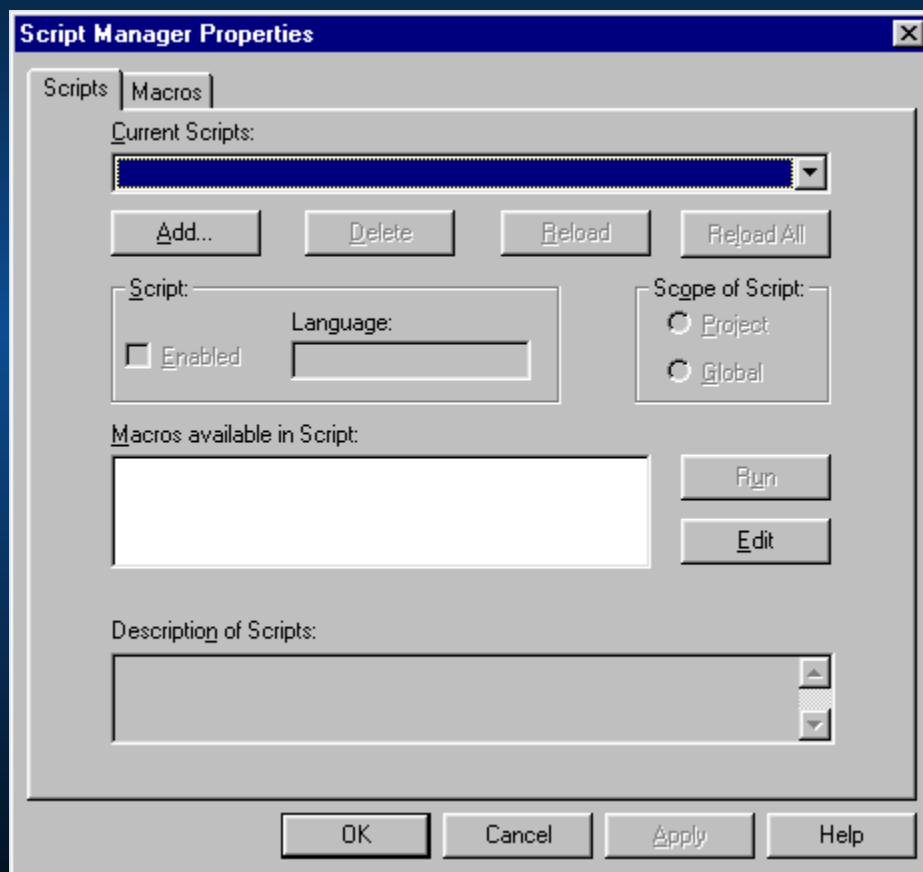


- All Pro-DG object commands available to place on toolbars and menus
- Menus can dock or float
- Your own script commands are also available

Customising The Keyboard



- Each IDE command can be bound to an accelerator
- Accelerators can be sequences of several keys
- Choose predefined maps or create/modify others
- BRIEF key map included!



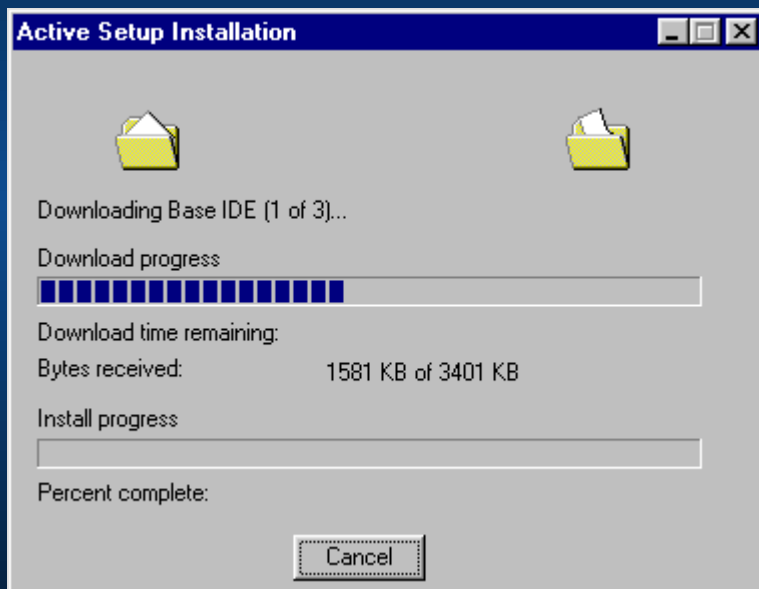
- Write many macros in a single script
- VBScript and JScript
- Full access to Pro-DG object model
- Scripts will be able to perform target comms in later versions

The Power Of DHTML

Document Title			
Category	Command	Keys	Description
Miscellaneous	CopyPaneToClipboard		Copy the pane and its layout to the clipboard
	PastePaneLink		Open the file in the clipboard in the current pane. Otherwise, paste the file/pane in the active pane
	PastePaneMove		Replace the currently active pane with the file or pane definition in the clipboard
	PastePaneOpen		Perform a general open using the file or pane definition on the clipboard
	PastePaneTarget		Starts a target pane operation with the pane in the clipboard
	PaneBack	Ctrl+I	Switch to the

- Dynamic HTML - create modal dialogs and write or modify HTML pages on the fly
- Used within Pro-DG for version reports
- Also to list all commands and accelerators

Active Setup



- Install and upgrade over HTTP or from CD
- Upgrades only the changed components

Existing Build Tools

- Build Manager components for compiler, assembler, linker
- Others (librarian, symmunge, dmpsx etc) to follow
- Any component can be called from command line

What About The Debugger?

- Currently must use existing Windows debugger
- Debugger and file serving to be rewritten into Pro-DG, starting asap
- Schedule: end of year :-)

Pick up your prescription now



SN Systems Software Ltd

Windows Programming Team:

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Duncan Lees, John Roberts (Lead), Tim Veal